

The MultiMediaCard

System Specification

Version 4.1

MMCA Technical Committee

Revision History

Version	Date	Section/ Page	Changes compared to previous issue
1.0	09-09-96	all	Initial version of system specification
1.1	27-09-96	Chap. 4,6,10	Complete revision
1.2	15-11-96	all	
1.4	25-2-98	all	Complete revision. Added new chapters
2.0	1/99	all	Several changes in SPI mode (chapter 7). Smaller changes in chapters 2, 4, 5. All references to “MMC” changed to “MultiMediaCard”.
2.1	5/99	4, 5, 7, 9, 10, 11	Several new features introduced: • Data interchange format • Password locking • Application specific commands Several clarifications and minor corrections for MultiMediaCard mode (chapter 4) and SPI mode (chapter 7). Backward compatible change of card dimensional tolerances (chapter 9).
2.11	6/99	4	Corrections in the status/command cross reference table (chapter 4)
2.2	1/00	7, 9	ESD tolerance, SPI status bits, Definition of card top side.
3.0	1/01	All	This is a major upgrade of the spec that contains new mechanical, electrical as well as protocol changes. See appendix for details.
3.1	6/01	5	The CSD version identification fields in Version 3.0 are not compatible with old versions. 3.1 fixes the issues, replaces and obsoletes 3.0
3.2	1/02	4, 5, 6, 9	Definition of Low Voltage MultiMediaCard completed. Some details added to the mechanical form factor of the card. Erase and Group write protect classes defined as mandatory for Read/Write cards and 2 CSD bits where allocated for application specific use.
3.3	3/03	2, 3, 8	Introduced the reduced size MMC. The high voltage range is now mandatory for all cards, as well as OCR register and CMD1 implementation. Corrected some typos in the wording.
3.31	5/03	A	Corrected connector order in Figure 73
4.0	12/03	All chapters	New features introduced are: high speed interface, 4 and 8 bus widths, power classes. The FAT16 file system is defined in a separate document, the File Formats chapter is no longer part of this document. The SWITCH command was adopted and a series of typos and clarifications were introduced. The most important being the error posting, which straightens up a lot of issues related to error/status bits. References to ‘Low voltage’ cards were changed to ‘Dual Voltage’

Revision History

Version	Date	Section/ Page	Changes compared to previous issue
4.1	01/05	All chapters	The major new items are the MMCmobile and MMCplus definitions including the minimum performance definition. Other significant additions are including mechanical notch and the friction force definition. Existing definitions for SPI mode and Stream command class were modified. Some typo corrections were also made. For the complete list of changes please see Appendix B10. Additional changes include: - Updating of all tables (of contents) and lists (of tables and figures) - Table 29 on page 80: added item to CSD_Structure register: Value 2 for versions 4.0 and 4.1 - Table 30 on page 80: added item to Spec_Vers register: Value 4 for version 4.1 - Table 48 on page 90: added new item to CSD_Structure register - placed Version 4.1 for Value 1.2; removed CSD_Structure register Value 3 - Table 49 on page 90: added new item to Extended CSD revision register: Value 1 for revision 1.1

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Table Of Contents

1	General Description	15
2	System Features	17
3	MultiMediaCard System Concept	19
3.1	Card Concept	22
3.1.1	MMC Plus and MMC Mobile	23
3.1.2	Form Factors	24
3.2	Bus Concept	24
3.2.1	Bus Lines	25
3.2.2	Bus Protocol	25
3.3	Controller Concept	30
3.3.1	Application Adapter Requirements	30
3.3.2	MultiMediaCard Adapter Architecture	31
4	MultiMediaCard Functional Description	33
4.1	General	33
4.2	Card Identification Mode	34
4.2.1	Card Reset	34
4.2.2	Operating Voltage Range Validation	34
4.2.3	Card Identification Process	36
4.3	Interrupt Mode	36
4.4	Data Transfer Mode	37
4.4.1	Command Sets And Extended Settings	39
4.4.2	High Speed Mode Selection	40
4.4.3	Power Class Selection	40
4.4.4	Bus Testing Procedure	41
4.4.5	Bus Width Selection	42
4.4.6	Data Read	42
4.4.7	Data Write	44
4.4.8	Erase	47
4.4.9	Write Protect Management	47
4.4.10	Card Lock/Unlock Operation	47
4.4.11	Application specific commands	50
4.5	Clock Control	51
4.6	Error Conditions	51
4.6.1	CRC and Illegal Command	51
4.6.2	Read, Write, Erase And Force Erase Time-out Conditions	52
4.6.3	Read ahead in Stream and multiple block read operation	52

4.7	Minimum Performance	52
4.7.1	Speed Class Definition	53
4.7.2	Absolute Minimum	53
4.7.3	Measurement of the Performance	53
4.8	Commands	54
4.8.1	Command Types	54
4.8.2	Command Format	54
4.8.3	Command Classes	54
4.8.4	Detailed Command Description	55
4.9	Card State Transition Table	61
4.10	Responses	62
4.11	Card Status	64
4.12	Memory Array Partitioning	68
4.13	Timings	69
4.13.1	Command and Response	70
4.13.2	Data Read	72
4.13.3	Data Write	73
4.13.4	Bus Test Procedure Timing	75
4.13.5	Timing Values.	76
5	Card Registers	77
5.1	OCR Register	77
5.2	CID Register	77
5.3	CSD Register	78
5.4	Extended CSD Register	86
5.5	RCA Register	91
5.6	DSR Register	91
6	SPI Mode	93
6.1	Introduction	93
6.2	SPI Interface Concept	93
6.3	SPI Bus Topology	93
6.4	MultiMediaCard Registers in SPI Mode	94
6.5	SPI Bus Protocol	95
6.5.1	Mode Selection	95
6.5.2	Bus Transfer Protection	96
6.5.3	Data Read	96
6.5.4	Data Write	98
6.5.5	Erase & Write Protect Management	99
6.5.6	Read CID/CSD Registers	100
6.5.7	Reset Sequence	100

6.5.8	Clock Control	100
6.5.9	Error Conditions	101
6.5.10	Memory Array Partitioning	102
6.5.11	Card Lock/unlock	102
6.5.12	Application Specific commands	102
6.6	SPI Mode Transaction Packets	102
6.6.1	Command Tokens	102
6.6.2	Responses	106
6.6.3	Data Tokens	109
6.6.4	Data Error Token	109
6.6.5	Clearing Status Bits	110
6.7	Card Registers	113
6.8	SPI Bus Timing Diagrams	113
6.8.1	Command / Response	114
6.8.2	Data read	115
6.8.3	Data write	116
6.8.4	Timing Values	117
6.9	SPI Electrical Interface	117
6.10	SPI Bus Operating Conditions	117
6.11	Bus Timing	117
7	Error protection	119
7.1	Error Correction Codes (ECC)	119
7.2	Cyclic Redundancy Codes (CRC)	119
8	MultiMediaCard Mechanical Specification	123
8.1	Card Package	123
8.1.1	External Signal Contacts (ESC)	123
8.1.2	Design and Format	124
8.1.3	Reliability and Durability	128
8.1.4	Quality Assurance	128
8.2	System: Card and Connector	128
8.2.1	Card Hot Insertion	129
8.2.2	Inverse Insertion	129
8.2.3	Card Orientation	129
9	The MultiMediaCard Bus	131
9.1	Hot Insertion and Removal	131
9.2	Power Protection	132
9.3	Power Up	133
9.4	Programmable Card Output Driver	134

9.5	Bus Operating Conditions	136
9.6	Bus Signal Levels	137
9.6.1	Open-Drain Mode Bus Signal Level	137
9.6.2	Push-Pull Mode Bus Signal Level - High Voltage MultiMediaCard	137
9.6.3	Push-Pull Mode Bus Signal Level - Dual voltage MultiMediaCard	138
9.7	Bus Timing	139
9.7.1	Card Interface Timings	139
10	MultiMediaCard Standard Compliance	141
11	File Formats For The MultiMediaCard	143
12	Abbreviations and Terms	145
Appendix A:	Application notes	147
A.1	Power Supply Decoupling	147
A.2	Payload Block Length and ECC Types Handling	147
A.3	Connector	147
A.3.1	General	148
A.3.2	Card Insertion and Removal	148
A.3.3	Characteristics	149
A.4	Description of method for storing passwords on the card	149
A.5	MultiMediaCard Macro Commands	150
A.6	Host Interface Timing	156
A.7	High Speed MultiMediaCard Bus Functions	157
A.7.1	Bus Initialization	157
A.7.2	Switching to High Speed Mode	158
A.7.3	Changing the Data Bus Width	158
Appendix B:	Changes Between System Specification Versions	161
B.1	Changes from Version 1.4 to 2.0	161
B.2	Changes from Version 2.0 to 2.11	161
B.3	Changes from Version 2.11 to 2.2	162
B.4	Version 3.0	162
B.5	Changes from Version 2.2 to 3.1	162
B.6	Changes from Version 3.1 to 3.2	163
B.7	Changes from Version 3.2 to 3.3	164
B.8	Changes from Version 3.3 to 3.31	164
B.9	Changes from Version 3.31 to 4.0	164
B.10	Changes from Version 4.0 to 4.1	165

List Of Tables

Table 1:	MMC System Operational Modes	18
Table 2:	MultiMediaCard Interface Pin Configuration.....	22
Table 3:	MultiMediaCard Registers	23
Table 4:	Bus Modes Overview	33
Table 5:	EXT_CSD Access Modes	40
Table 6:	1-bit Bus Testing Pattern.....	41
Table 7:	4-bit Bus Testing Pattern.....	42
Table 8:	8-bit Bus Testing Pattern.....	42
Table 9:	Lock Card Data Structure	48
Table 10:	Command Code Length	54
Table 11:	Card Command Classes (CCCs)	55
Table 12:	Basic Commands And Read Stream Commands (Class 0 And Class 1)	56
Table 13:	Block Oriented Read Commands (Class 2)	57
Table 14:	Stream Write Commands (Class 3).....	57
Table 15:	Block Oriented Write Commands (Class 4)	58
Table 16:	Block Oriented Write Protection Commands (Class 6)	58
Table 17:	Erase Commands (Class 5)	59
Table 18:	I/O Mode Commands (Class 9)	59
Table 19:	Lock Card (Class 7)	59
Table 20:	Application Specific Commands (Class 8)	60
Table 21:	Card State Transition Table	61
Table 22:	Card Status	65
Table 23:	Card Status Field/Command - Cross Reference	67
Table 24:	Timing Diagram Symbols.....	69
Table 25:	Timing Parameters	76
Table 26:	OCR Register Definition.....	77
Table 27:	The CID fields.....	77
Table 28:	The CSD fields.....	79
Table 29:	CSD Register Structure.....	80
Table 30:	System Specification Version	80
Table 31:	TAAC Access Time Definition.....	80
Table 32:	Maximum Bus Clock Frequency Definition.....	81
Table 33:	Supported Card Command Classes.....	81
Table 34:	Data Block Length	81
Table 35:	DSR Implementation code Table.....	82
Table 36:	VDD,min Current Consumption	82
Table 37:	VDD,max Current Consumption	83
Table 38:	Multiply Factor For The Device Size	83
Table 39:	R2W_FACTOR	84
Table 40:	File Formats	85
Table 41:	ECC Type.....	85
Table 42:	CSD Field Command Classes	86
Table 43:	Extended CSD	87

Table 44:	Card Supported Command Sets	88
Table 45:	R/W Access Performance Values	88
Table 46:	Power Classes.....	89
Table 47:	Card Type	90
Table 48:	CSD Register Structure	90
Table 49:	Extended CSD Revision.....	90
Table 50:	Standard MMC Command Set Revisions	90
Table 51:	Power Class	91
Table 52:	Bus Mode Value	91
Table 53:	SPI Interface Pin Configuration	94
Table 54:	MultiMediaCard Registers In SPI Mode.....	95
Table 55:	Command Classes In SPI Mode.....	103
Table 56:	Commands And Arguments	104
Table 57:	Status Bit Descriptions	112
Table 58:	SPI Status Field/Command - Cross Reference.....	112
Table 59:	SPI Timing Values	117
Table 60:	DSR Register Contents.....	134
Table 61:	High Speed Card Interface Timing	139
Table 62:	Backwards Compatible Card Interface Timing.....	140
Table 63:	Macro Commands	150
Table 64:	Forwards Compatible Host Interface Timing.....	156
Table 65:	Test Pattern for 8 Bit Bus.....	159
Table 66:	Test Pattern for 4 Bit Bus.....	159
Table 67:	Test Pattern for 1 Bit Bus	159

List Of Figures

Figure 1:	Topology Of MultiMediaCard Systems.....	19
Figure 2:	MultiMediaCard System Overview	20
Figure 3:	MultiMediaCard System Example.....	21
Figure 4:	MultiMediaCard Architecture.....	24
Figure 5:	MultiMediaCard Form Factors (Bottom View).....	24
Figure 6:	MultiMediaCard Bus System.....	25
Figure 7:	Sequential Read Operation.....	26
Figure 8:	(Multiple) Block Read Operation	26
Figure 9:	Sequential Write Operation.....	27
Figure 10:	(Multiple) Block Write Operation.....	27
Figure 11:	“no response” And “no data” Operations	27
Figure 12:	Command Token Format	28
Figure 13:	Response Token Format.....	28
Figure 14:	Data Packet Format.....	29
Figure 15:	MultiMediaCard Controller Scheme.....	30
Figure 16:	MultiMediaCard Adaptor Architecture.....	31
Figure 17:	MultiMediaCard State Diagram (Card Identification Mode)	35
Figure 18:	MultiMediaCard State Transition Diagram, Interrupt Mode.....	37
Figure 19:	MultiMediaCard State Diagram (Data Transfer Mode).....	38
Figure 20:	Memory Array Partitioning.....	69
Figure 21:	Identification Timing (Card Identification Mode).....	70
Figure 22:	SET_RCA Timing (Card Identification Mode)	70
Figure 23:	Command Response Timing (Data Transfer Mode).....	71
Figure 24:	R1b Response Timing.....	71
Figure 25:	Timing Response End To Next Command Start (Data Transfer Mode).....	71
Figure 26:	Timing Of Command Sequences (All Modes)	71
Figure 27:	Single Block Read Timing	72
Figure 28:	Multiple Block Read Timing	72
Figure 29:	Stop Command Timing (CMD12, Data Transfer Mode).....	72
Figure 30:	Block Write Command Timing.....	73
Figure 31:	Multiple Block Write Timing	74
Figure 32:	Stop Transmission During Data Transfer From The Host.....	74
Figure 33:	Stop Transmission During CRC Status Transfer From The Card	74
Figure 34:	Stop Transmission After Last Data Block. Card Is Busy Programming.	75
Figure 35:	Stop Transmission After Last Data Block. Card Becomes Busy.....	75
Figure 36:	4-bit System Bus Testing Procedure.....	76
Figure 37:	MultiMediaCard Bus System.....	94
Figure 38:	SPI Single Block Read Operation	96
Figure 39:	SPI Multiple Block Read Operation	96
Figure 40:	SPI Read Operation - Data Error	97
Figure 41:	SPI Single Block Write Operation	98
Figure 42:	SPI Multiple Block Write Operation	98
Figure 43:	SPI ‘No data’ Operations	100

Figure 44:	R1 Response Format	107
Figure 45:	R1b Response Format	107
Figure 46:	R2 Response Format	108
Figure 47:	R3 Response Format	108
Figure 48:	Data Response Format	109
Figure 49:	Start Data Block Token Format	109
Figure 50:	Data Error Token.....	110
Figure 51:	SPI Command/Response Transaction, Card Is Ready	114
Figure 52:	SPI Command/Response Transaction, Card Is Busy	114
Figure 53:	SPI Card Response To The Next Host Command	114
Figure 54:	SPI Single Block Read	115
Figure 55:	SPI Multiple Block Read, Stop Transmission Does Not Overlap Data	115
Figure 56:	SPI Multiple Block Read, Stop Transmission Overlaps Data.....	115
Figure 57:	SPI Read CSD and CID Registers.....	116
Figure 58:	SPI Single Block Write	116
Figure 59:	SPI Multiple Block Write.....	116
Figure 60:	CRC7 Generator/Checker	120
Figure 61:	CRC16 Generator/Checker	121
Figure 62:	Dimensions Of A Normal Size MultiMediaCard (Bottom View, DIN)	125
Figure 63:	Dimensions Of A Reduced Size MultiMediaCard (DIN)	126
Figure 64:	Location Of Pads' Via Holes	126
Figure 65:	Conductive Material Usage Restrictions.....	127
Figure 66:	Hold-back notch tolerances	127
Figure 67:	Inverse Insertion	129
Figure 68:	Bus Circuitry Diagram	131
Figure 69:	Improper Power Supply	132
Figure 70:	Shortcut Protection	132
Figure 71:	Power-up Diagram	133
Figure 72:	MultiMediaCard Bus Driver	135
Figure 73:	Bus Signal Levels.....	137
Figure 74:	Timing Diagram: Data Input/Output.....	139
Figure 75:	Power Supply Decoupling.....	147
Figure 76:	Modified MultiMediaCard Connector For Hot Insertion.....	148
Figure 77:	Legend For Command Sequences' Flow Charts.....	151
Figure 78:	CIM_SINGLE_CARD_ACQ.....	152
Figure 79:	CIM_SETUP_CARD	152
Figure 80:	CIM_STREAM_READ	153
Figure 81:	CIM_READ_BLOCK.....	154
Figure 82:	CIM_READ_MBLOCK	154
Figure 83:	CIM_WRITE_MBLOCK.....	155
Figure 84:	CIM_ERASE_GROUP	156

1 General Description

The MultiMediaCard is an universal low cost data storage and communication media. It is designed to cover a wide area of applications as smart phones, cameras, organizers, PDAs, digital recorders, MP3 players, pagers, electronic toys, etc. Targeted features are high mobility and high performance at a low cost price. These features include low power consumption and high data throughput at the memory card interface.

The MultiMediaCard communication is based on an advanced 13-pin bus. The communication protocol is defined as a part of this standard and referred to as the MultiMediaCard mode. To ensure compatibility with existing controllers, the cards may offer, in addition to the MultiMediaCard mode, an alternate communication protocol which is based on the SPI standard.

To provide for the forecasted migration of CMOS power (V_{DD}) requirements and for compatibility and integrity of MultiMediaCard systems, two types of MultiMediaCards are defined in this standard specification, which differ only in the valid range of system V_{DD} . These two card types are referred to as High Voltage MultiMediaCard and Dual Voltage MultiMediaCard.

The purpose of the system specification is the definition of the MultiMediaCard, its environment and handling. It gives guidelines for a system designer. The system specification also defines a tool box (a set of macro functions and algorithms) which contributes to reducing the design-in costs.

The document is split up into several portions. The MultimediaCard Features are described in [Chapter 2](#).

[Chapter 3](#) gives a general overview of the system components: card, bus, and host.

The common MultiMediaCard characteristics are described in [Chapter 4](#). As this description defines an overall set of card properties, you should work with the vendor-specific, product documentation in parallel.

[Chapter 5](#) describes the card registers.

The SPI mode is described in [Chapter 6](#).

All error protection techniques employed in this standard are described in [Chapter 7](#).

[Chapter 8](#) describes the physical and mechanical properties of the cards and the minimal requirements of the card slots and cartridges.

[Chapter 9](#) defines the MultiMediaCard bus as a universal communication interface and the electrical parameters of the interface.

The standard compliance criteria for the cards and hosts are described in [Chapter 10](#).

For achieving high data interchangeability, three basic file formats are defined in [Chapter 11](#) as valid file formats for the MultiMediaCard

[Appendix A](#) contains additional information that is informative in nature and not considered a constituent part of this specification. These Application Notes contain useful hints for the circuit and system designers, helping simplify the design process.

[Appendix B](#) lists the major changes between the previous and the current version of this specification.

As used in this document, “shall” or “will” denotes a mandatory provision of the standard. “Should” denotes a provision that is recommended but not mandatory. “May” denotes a feature whose presence does not preclude compliance, that may or may not be present at the option of the implementor.

2 System Features

The MultiMediaCard System has a wide variety of system features, whose comprehensive elements serves several purposes, which include:

- Covering a broad category of applications from smart phones and PDAs to digital recorders and toys
- Facilitating the work of designers who seek to develop applications with their own advanced and enhanced features
- Maintaining compatibility and compliance with current electronic, communication, data and error handling standards.

The following list identifies the main features of the MultiMediaCard System, which:

- **Is targeted for portable and stationary applications**
- **Has these System Voltage (V_{DD}) Ranges:**

	High Voltage MultiMediaCard	Dual Voltage MultiMediaCard
Communication	2.7 - 3.6	1.65 - 1.95, 2.7 - 3.6¹
Memory Access	2.7 - 3.6	1.65 - 1.95, 2.7 - 3.6

1) V_{DD} range: 1.95V - 2.7V is not supported.

- **Includes MMCplus and MMCmobile definitions**
- **Is designed for read-only, read/write and I/O cards**
- **Supports card clock frequencies of 0-20MHz, 0-26MHz or 0-52MHz**
- **Has a maximum data rate up to 416Mbits/sec.**
- **Has a defined minimum performance**
- **Maintains card support for three different data bus width modes: 1bit (default), 4bit and 8bit**
- **Includes password protection of data**
- **Supports basic file formats for high data interchangeability**
- **Includes application specific commands**
- **Enables correction of memory field errors**
- **Has built-in write protection features, which may be permanent or temporary**
- **Includes a simple erase mechanism**
- **Maintains full backward compatibility with previous MultiMediaCard systems (1 bit data bus, multi-card systems)**
- **Ensures that new hosts retain full compatibility with previous versions of MultiMediaCards (backward compatibility).**

- **Supports two form factors: Normal size(24mm x 32mm x 1.4mm) and Reduced size (24mm x 18mm x 1.4mm)**
- **Supports multiple command sets**
- **Includes attributes of the available operation modes:**

MultiMediaCard Mode	SPI Mode
Ten-wire bus (clock, 1 bit command, 8 bit data bus)	Three-wire serial data bus (clock, dataIn, dataOut) + card specific CS signal.
Card selection is done through an assigned unique card address to maintain backwards compatibility to prior versions of the specification	Card selection via a hardware CS signal
One card per MultiMediaCard bus	Card requires a dedicated CS signal.
Easy identification and assignment of session address	Not available. Card selection via a hardware CS signal
Error-protected data transfer	Optional. A non-protected data transfer mode is available.
Sequential and Single/Multiple block Read/Write commands	Single/Multiple block Read/Write commands

Table 1: MMC System Operational Modes